VIETNAMESE - GERMAN UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE



# Customer Segmentation Analysis Report

## Customer and Segmentation Data Analytics

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| Nguyễn Hà Khải | Dimensions implementation, data warehouse design | 100% |
| Nguyễn Quang Ngọc Trác | K-means clustering, RFM technique exploration implementaion | 100% |
| Tô Huỳnh Phúc | Fact tables, schedulers, and incremental loading implementation | 100% |
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# Chapter 1

# Introduction

## 1.1 Project Context and Data Source

In modern retail, analyzing customer purchasing behavior is critical for making effective business decisions. Traditional transaction tables store raw order data, but they do not automatically reveal which customers purchase repeatedly, which generate the highest value, or which groups require targeted marketing strategies.

This project analyzes **CompanyX**, a global retailer specializing in bicycles, accessories, and apparel. The transactional database records customer demographics, sales orders, product lines, and order details from May 2011 through June 2014. To provide a consistent baseline for analysis, 30 June 2014 is selected as the primary historical cutoff date for the reporting dashboards.

The project focuses on *customer-centric analytics* rather than basic sales reporting. Specifically, our solution combines Recency, Frequency, and Monetary (RFM) modeling, unsupervised K-Means clustering, and a dimensional data warehouse to segment customers into actionable groups. An automated ETL pipeline keeps customer profiles up to date and re-clusters buyers near-real-time after new transactions, while also storing monthly snapshots to track long-term behavioral trends.

## 1.2 Business Problem

Retail managers need to know not only overall sales figures, but also *which customer groups drive revenue and how their purchasing habits evolve over time.* Four key business challenges motivated this project:

- **Customer Retention and Inactivity:** Identifying one-time buyers versus loyal repeat customers is essential. Without tracking purchase frequency and the elapsed time since

the last order, marketing teams cannot easily detect customer attrition.

- **Uneven Spending Distribution:** A small proportion of high-value buyers often accounts for the majority of company revenue. Treating all customers identically leads to wasted marketing budgets and missed VIP loyalty opportunities.
- **Cross-Selling and Up-Selling Opportunities:** Understanding a customer's spending tier and engagement stage helps the sales team recommend complementary items, such as bike accessories and apparel, at the right moment.
- **Operational vs. Historical Analysis:** An operational view updated immediately after a purchase answers different questions than a standardized month-end snapshot. Both views are required to manage immediate transaction events while tracking multi-year customer migration.

## 1.3 Objectives and Deliverables

The project delivers an end-to-end Business Intelligence solution that combines **data mining customer segmentation** with an **automated data pipeline**. The core technical objectives are:

1. Design and implement a customer-focused dimensional data warehouse featuring Type 0, Type 1, and Type 2 Slowly Changing Dimensions (SCD);
2. Build an automated SSIS ETL pipeline with incremental change detection to load customer profiles efficiently;
3. Implement a near-real-time operational scoring path that recalculates RFM metrics and assigns clusters following new purchases;
4. Construct a monthly periodic snapshot process to capture historical customer behavior and support trend analytics;
5. Train, validate, and version K-Means clustering models in Python, persisting cluster centroids and scaling parameters in the warehouse;
6. Deliver an interactive, multi-page Power BI dashboard that translates analytical findings into practical business strategies.

The delivered artifacts include the SQL Server warehouse schema, deployed SSIS ETL packages, Python machine learning scripts, an Apache Airflow retraining DAG, and a comprehensive Power BI dashboard.

## 1.4 Report Organization

The remainder of this report is structured as follows:

- Chapter 2 reviews theoretical concepts, RFM methodology, and comparable industry applications;
- Chapter 3 details the dimensional warehouse architecture and SCD implementations;
- Chapter 4 presents the operational ETL workflow, incremental data flows, and scheduling;
- Chapter 5 explains the RFM feature engineering, data transformations, K-Means clustering, and cluster naming;
- Chapter 6 walks through the three-page interactive Power BI dashboard and core DAX measures;
- Chapter 7 translates data findings into actionable marketing recommendations and a decision-support use case;
- Chapter 8 describes system integration, verification checks, and engineering boundaries;
- Chapter 9 summarizes project achievements and outlines directions for future enhancement.

# Chapter 2

# Background and Related Applications

## 2.1 Customer Segmentation with RFM and K-Means

Recency, Frequency, and Monetary (RFM) analysis is a widely adopted technique in direct marketing and retail analytics. It summarizes purchasing behavior into three measurable dimensions:

- **Recency ($R$):** Number of days elapsed between a reference evaluation date and the customer's most recent order. Lower values indicate active, engaged shoppers.
- **Frequency ($F$):** Total number of distinct orders placed by the customer within a defined time window (specifically a trailing 365-day period in this project).
- **Monetary ($M$):** Total financial spend accumulated across all order lines within the same trailing 365-day window.

By evaluating actual purchasing transactions rather than static demographic categories, RFM modeling provides an empirical baseline for customer valuation and engagement tracking.

In traditional retail practice, RFM segmentation relies on heuristic rules, such as manually dividing each metric into quintiles (1 to 5) or hardcoded SQL thresholds. While intuitive, rule-based binning requires arbitrary boundary choices and treats each variable independently. In contrast, **K-Means clustering** groups customers algorithmically based on Euclidean distance in a normalized multi-dimensional feature space. By applying logarithmic transformations and standardization, K-Means discovers natural groupings from historical transaction patterns without predefined thresholds.

To make machine learning segmentation operationally viable, the model must be embedded within an integrated data architecture. A **dimensional data warehouse** provides durable customer surrogate keys, tracks demographic shifts, stores periodic historical snapshots, and

persists trained model parameters. Connecting this warehouse to **Power BI** dashboards allows managers to interactively explore segment distributions and monitor business KPIs.

## 2.2 Related Industry Applications and Best Practices

### 2.2.1 Black Diamond: Data-Driven Customer Retention

In the outdoor retail sector, equipment manufacturer Black Diamond utilized customer data analytics to unify fragmented transaction records and deploy targeted retention programs (Lexer 2019). By separating active high-value riders from lapsed buyers, the company implemented automated re-engagement workflows that dramatically reduced customer acquisition costs and boosted email campaign revenue. This demonstrates the tangible commercial value of identifying inactive, high-potential accounts before permanent churn occurs.

### 2.2.2 K-Means Clustering in Retail Marketing

Academic and industry tutorials highlight K-Means as an effective unsupervised technique for behavioral segmentation (Towards AI 2022). These studies emphasize that raw retail data cannot simply be fed into clustering algorithms directly; rigorous preprocessing—including outlier mitigation, feature scaling, and cluster validation (e.g., Elbow and Silhouette diagnostics)—is essential to generate statistically cohesive and commercially interpretable segments.

### 2.2.3 Alibaba Cloud: End-to-End Pipeline Operationalization

Modern enterprise analytics requires moving beyond isolated, one-off machine learning experiments in desktop notebooks. Pratama demonstrated an end-to-end architecture on Alibaba Cloud integrating cloud storage, automated feature engineering, K-Means model training, and operational scoring (Pratama 2020). For CompanyX, this highlights the necessity of combining SSIS automated loading with Python scoring scripts and database parameter persistence, ensuring customer segments update reliably as new transactions occur.

## 2.3 Comparison of Methodological Approaches

Table 2.1 summarizes the strengths and limitations of the analytical approaches considered for this project.

Table 2.1: Comparison of customer segmentation and data pipeline approaches.

| **Approach** | **Typical Implementation** | **Relevance and Limitations** |
|---|---|---|
| Rule-Based RFM | Hardcoded SQL thresholds or quintile ranking | Highly transparent and simple to compute, but requires arbitrary boundary selection and ignores multi-variable interactions. |
| Standalone K-Means Clustering | Unsupervised clustering in Jupyter / Python notebooks | Effectively uncovers natural clusters from data, but lacks automated data integration, historical versioning, and operational deployment. |
| Integrated BI Solution (CompanyX) | Dimensional Warehouse + SSIS ETL + Python K-Means + Power BI | Combines automated near-real-time scoring with longitudinal snapshot reporting and interactive decision-support dashboards. |

## 2.4 Problem Statement and Chosen Solution

To maximize commercial competitiveness, CompanyX needs clear answers to three fundamental business questions:

1. *Which customer segments drive the vast majority of company revenue?*
2. *Which high-value buyers are exhibiting signs of disengagement and require urgent retention?*
3. *How should marketing budgets be allocated across distinct demographic and behavioral personas?*

Our solution addresses these questions through an end-to-end architecture featuring two complementary fact tables (near-real-time purchase scoring and monthly historical snapshots), a five-cluster K-Means model, and an executive Power BI dashboard designed for actionable decision-making.

# Chapter 3

# Data Warehouse Design

The CompanyX data warehouse employs a **fact constellation (galaxy) schema** centered on customer analytics. Rather than modeling transactional order lines, the warehouse establishes the *individual customer* as the core analytical grain. Two complementary fact tables share standardized dimensions for customer demographics, socioeconomic profiles, machine learning model versions, and calendar dates, as illustrated in Figure 3.1.

## 3.1 Staging Layer

The staging table `stg.CustomerProfile` serves as a temporary landing zone during ETL extraction. It uses a composite primary key consisting of a unique batch identifier (`BatchID`) and the source customer key (`CustomerKey`). Each execution of the SSIS customer package generates a new GUID for `BatchID`, truncates prior staged records, extracts changed demographic rows from the source database, and executes the merge procedure `usp_ApplyCustomerProfileBatch`. This batch-scoped staging design ensures atomic transactions and prevents partial data loads from corrupting dimension tables.

## 3.2 Dimension Layer and Slowly Changing Dimensions

### 3.2.1 DimCustomerProfiling (SCD Type 0 and Type 1)

`DimCustomerProfiling` maintains one current demographic record per customer, keyed by `CustomerKey_Str` (converted from the source `CustomerID`). Attributes in this dimension are managed under two SCD strategies:

- **SCD Type 0 (Fixed):** Attributes such as `BirthDate` are immutable once inserted and never modified.

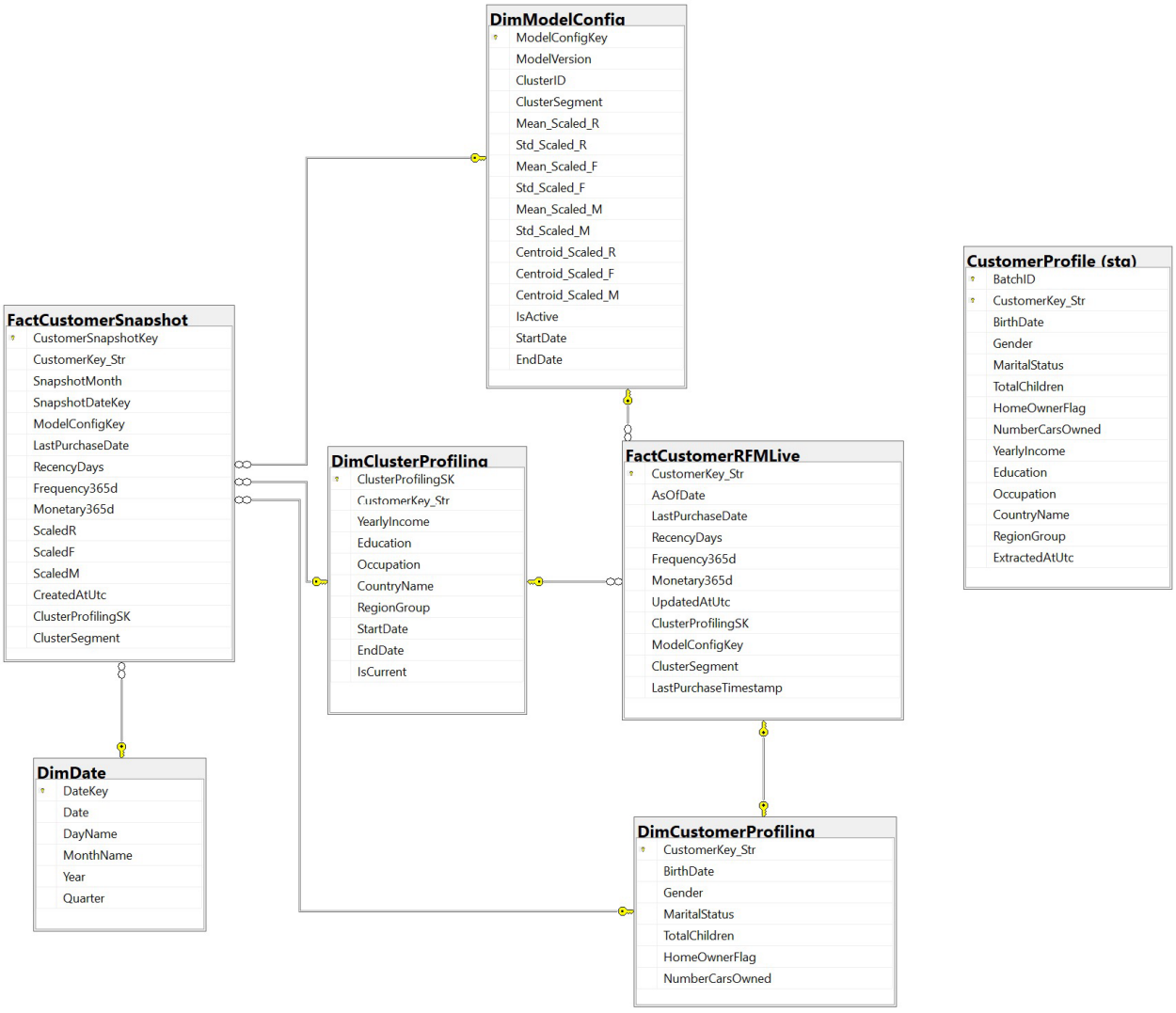



Figure 3.1: Final CompanyX customer-oriented fact constellation.

- **SCD Type 1 (Overwrite):** Personal attributes that change over time—including `Gender`, `MaritalStatus`, `TotalChildren`, `HomeOwnerFlag`, and `NumberCarsOwned`—are updated directly in place. This provides instant access to current customer statuses without multiplying dimension rows.

### 3.2.2 DimClusterProfiling (SCD Type 2)

`DimClusterProfiling` stores evolving socioeconomic and regional characteristics, including yearly income, education level, occupation, country, and region group.

To preserve historical context, these attributes are tracked using **SCD Type 2**:

- When a demographic change is detected, the existing active row is closed by setting its `EffectiveEndDate` to the current date and `IsCurrent = 0`.
- A new record is inserted with a unique surrogate key (`ClusterProfilingSK`), effective date intervals, and `IsCurrent = 1`.

This mechanism enables historical RFM snapshots to be accurately linked with the exact socioeconomic profile the customer possessed at that point in time.

### 3.2.3 DimModelConfig (ML Model Versioning)

`DimModelConfig` operationalizes machine learning by persisting trained K-Means configurations directly in the relational database. For each model version, five rows are stored (one per cluster), recording:

- Cluster ID (0 to 4) and human-interpretable business segment name;
- Feature scaling parameters (mean and standard deviation for Recency, Frequency, and Monetary);
- Centroid coordinates in standardized 3D feature space;
- Validity dates (`StartDate`, `EndDate`) and an active status flag (`IsActive`).

Persisting centroids and scaler parameters allows downstream scoring scripts to assign incoming customers to clusters instantly using matrix multiplication, eliminating the need to re-fit K-Means on every transaction.

### 3.2.4 DimDate

`DimDate` provides a standard enterprise calendar dimension with attributes including calendar year, quarter, month, day, and day of week. The initialization script pre-populates dates from **1 January 2010** through twenty years beyond execution, comfortably covering the source transaction history (2011–2014) and future operational reporting horizons.

## 3.3 Fact Layer

### 3.3.1 FactCustomerSnapshot (Monthly Periodic Snapshot)

`FactCustomerSnapshot` captures the behavioral state of every active customer at standardized monthly intervals. Its analytical grain is *one customer at one calendar month-end date*, uniquely identified by `CustomerKey_Str` and `SnapshotDateKey`.

At each month-end, the snapshot procedure reconstructs customer metrics from raw transactions:

- `Frequency365d`: Distinct sales orders placed within the trailing 365 days;
- `Monetary365d`: Total order line revenue within the trailing 365 days;
- `RecencyDays`: Elapsed days since the customer's most recent purchase (can exceed 365 days).

Scheduled to run at 00:05 on the first day of each month, this table provides the foundation for longitudinal trend analysis, customer churn tracking, and annual model retraining.

### 3.3.2 FactCustomerRFMLive (Purchase-Triggered Operational Fact)

`FactCustomerRFMLive` delivers near-real-time operational intelligence. Instead of logging every order line, it maintains exactly one row per customer reflecting their state immediately after their latest transaction.

When new sales are committed in the source database, the ETL process recalculates the affected customer's trailing 365-day frequency and monetary spend, while setting purchase-time recency to zero. The Python scoring script then maps the standardized vector against `DimModelConfig` to update the customer's cluster assignment.

By separating purchase-event operational scoring (`FactCustomerRFMLive`) from periodic historical observation (`FactCustomerSnapshot`), the warehouse achieves both high-frequency operational responsiveness and comprehensive retrospective reporting without unnecessary computational overhead.

# Chapter 4

# ETL Pipeline and Automation

## 4.1 The General View

Figure 4.1 summarizes the two complementary data paths and annual model training. The customer-demographics load and purchase-triggered RFM path support operational decisions; the monthly snapshot path preserves the state of customers at a common evaluation date.

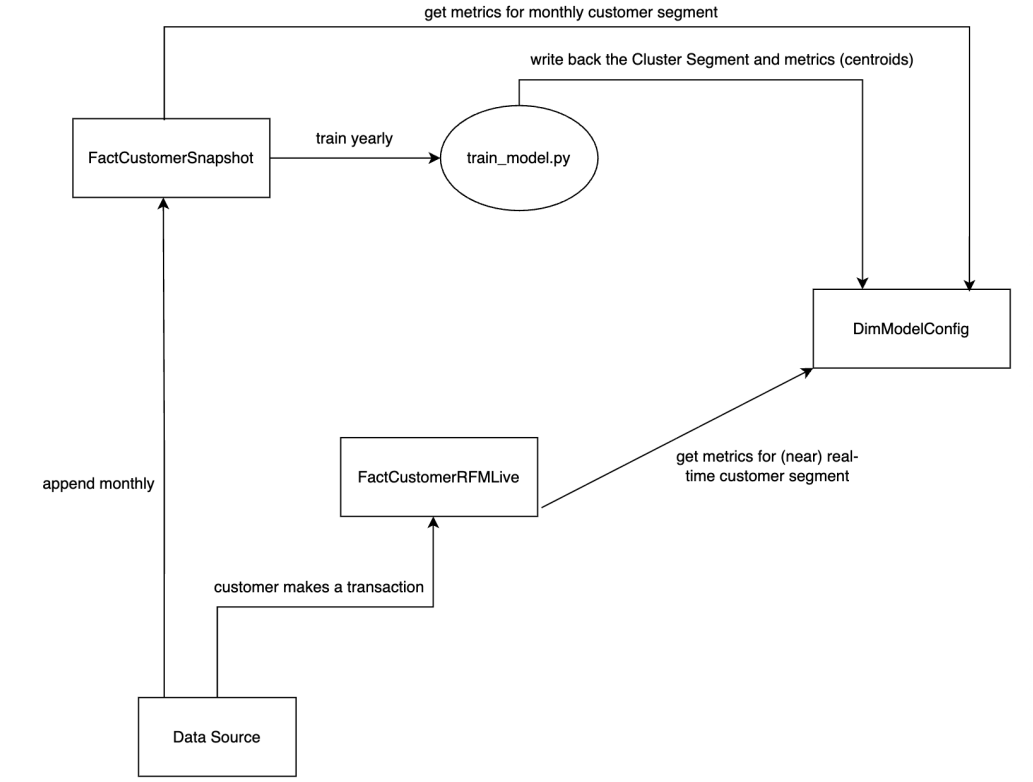



Figure 4.1: High-level flow from CompanyX transactions to RFM facts, model configurations, and reporting.

The first path processes customer transactions operationally. Deployed SSIS packages execute every minute to detect newly modified customer profiles and calculate purchase-time RFM metrics for affected buyers in **FactCustomerRFMLive**. A Python scoring script (`live_clustering.py`) applies the active parameters from `DimModelConfig`, standardizes the features, and assigns the customer to the closest cluster centroid in standardized RFM space. Because this operational view evaluates metrics at the customer's latest purchase event, it provides an immediate post-transaction classification.

The second path executes on a monthly schedule (00:05 on the first day of each month) to compute RFM states as of the *previous month-end date* and appends them to **FactCustomer-Snapshot**. This monthly table tracks customer dormancy over time and serves as the training source for annual K-Means model retraining.

For model initialization, `bootstrap.py` established the baseline 2011 model from year-end data, followed by annual retraining models for 2012 and 2013 using `train_model.py`. The Power BI dashboard presented in Chapter 6 utilizes the 30 June 2014 snapshot as its primary reporting reference.

## 4.2 ETL Pipeline & Data Flow

This section follows the three SSIS packages supplied in the project: `Package_Customer_Incremental.dtsx`, `RFM_Live.dtsx`, and `RFM_Snapshot.dtsx`. The following screenshots and SQL excerpts are from the team's implementation; full routine definitions are included in the system resources. The customer and live packages have been deployed on a one-minute schedule, while the snapshot package is scheduled monthly.

### 4.2.1 Customer Control Flow

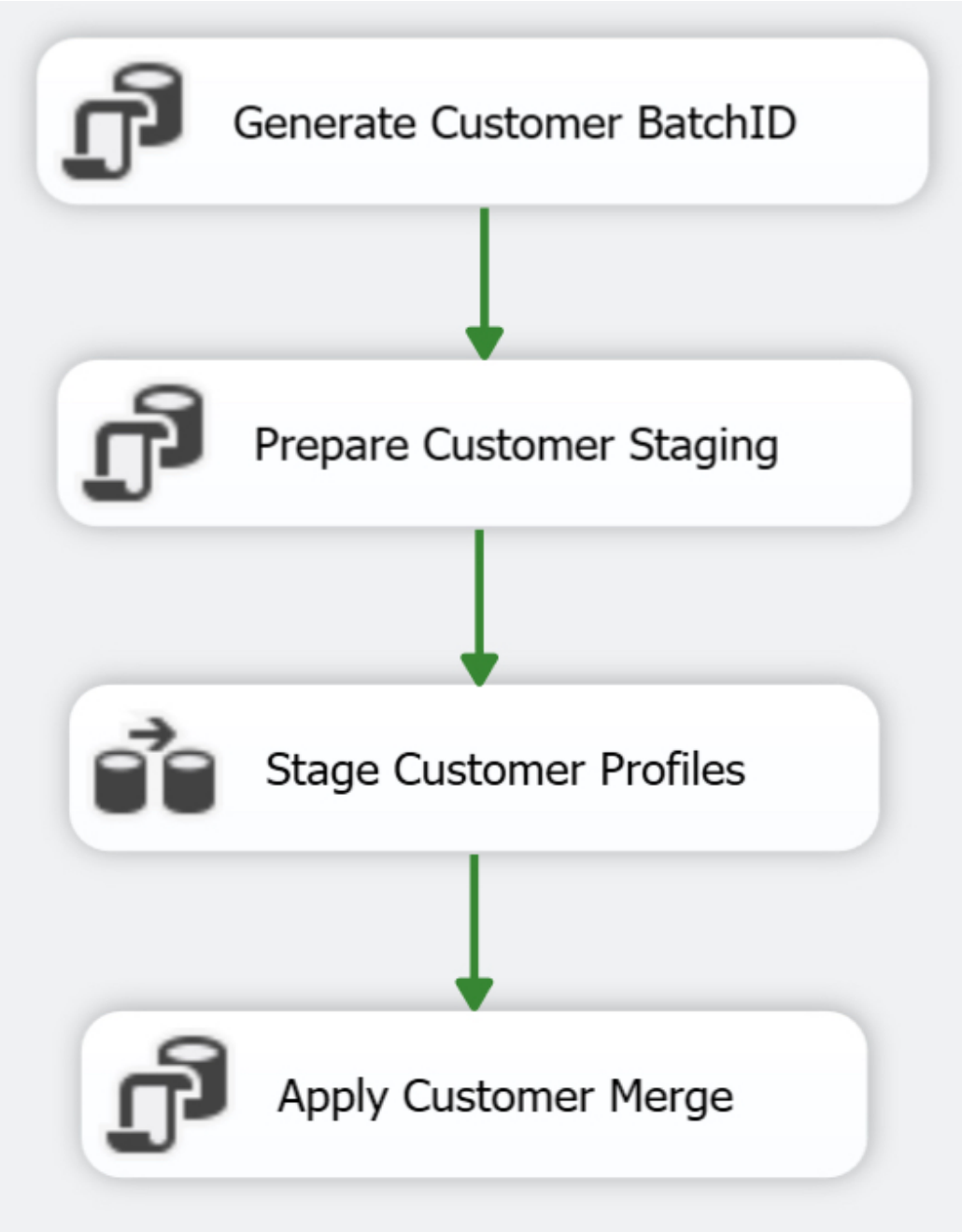



Figure 4.2: DimClusterProfiling and DimCustomerProfiling handling.

In the above figure, we show the overview of DimClusterProfiling and DimCustomerProfiling creation. First, a SQL task is executed to generate a new batch identifier:

```
SELECT CONVERT(VARCHAR(36), NEWID()) AS BatchID;
```

Listing 4.1: Batch ID Generation

After that, we clean the contents in previous runs in Preparing Customer Staging:

```
TRUNCATE TABLE CompanyX_DataWarehouse.stg.CustomerProfile;
```

Listing 4.2: Customer Profile Truncation

Following up, we examine the data flow inside Stage Customer Profiles:

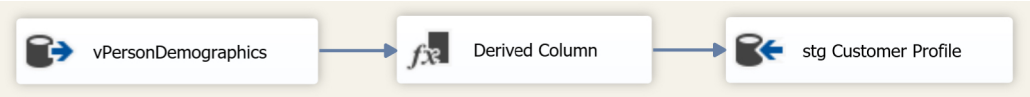



Figure 4.3: Data Flow of Stage Customer Profiles.

Overall, there are three primary components in this data flow:

- **vPersonDemographics (OLE DB Source):** Extracts changed demographic records from the CompanyX operational database:

```
SELECT
    c.CustomerID AS CustomerKey,
    d.BusinessEntityID,
    CAST(d.BirthDate AS DATE) AS BirthDate,
    d.Gender,
    d.MaritalStatus,
    d.YearlyIncome,
    d.TotalChildren,
    d.Education,
    d.Occupation,
    CAST(d.HomeOwnerFlag AS TINYINT) AS HomeOwnerFlag,
    d.NumberCarsOwned,
    CAST(d.DateFirstPurchase AS DATE) AS DateFirstPurchase,
    st.[Name] AS CountryName,
    st.[Group] AS RegionGroup
FROM Sales.vPersonDemographics AS d
INNER JOIN Sales.Customer AS c
    ON d.BusinessEntityID = c.PersonID
LEFT JOIN Sales.SalesTerritory AS st
    ON c.TerritoryID = st.TerritoryID
LEFT JOIN CompanyX_DataWarehouse.dbo.DimCustomerProfiling AS dc
    ON dc.CustomerKey_Str = CONVERT(NVARCHAR(20), c.CustomerID)
LEFT JOIN CompanyX_DataWarehouse.dbo.DimClusterProfiling AS dp
    ON dp.CustomerKey_Str = CONVERT(NVARCHAR(20), c.CustomerID)
   AND dp.IsCurrent = 1
WHERE d.DateFirstPurchase IS NOT NULL
```

```
27  AND (
28       dc.CustomerKey_Str IS NULL
29       OR dp.CustomerKey_Str IS NULL
30       OR EXISTS (
31            SELECT d.Gender, d.MaritalStatus, d.TotalChildren,
32                   CONVERT(BIT, d.HomeOwnerFlag), d.
    NumberCarsOwned
33            EXCEPT
34            SELECT dc.Gender, dc.MaritalStatus, dc.TotalChildren,
35                   dc.HomeOwnerFlag, dc.NumberCarsOwned
36       )
37       OR EXISTS (
38            SELECT d.YearlyIncome, d.Education, d.Occupation,
39                   st.[Name], st.[Group]
40            EXCEPT
41            SELECT dp.YearlyIncome, dp.Education, dp.Occupation,
42                   dp.CountryName, dp.RegionGroup
43       )
44  );
45
```

Listing 4.3: Table Selection

The extraction selects customers with a first purchase whose relevant demographic attributes have changed compared to the warehouse. The `EXCEPT` predicates compare groups of attributes, including nullable values, performing change detection without requiring a complete reload on every execution.

- **Derived Column Transformation:** Normalizes input values and standardizes missing demographic attributes before staging, assigning configured fallback defaults (such as `"Unknown"`, `0`, and `"U"`) as shown in Figure 4.4.

- **stg Customer Profile (OLE DB Destination):** Writes the transformed records directly into the staging table (`stg.CustomerProfile`, described in Section 3.1) ready for the subsequent T-SQL merge.

Finally, the *Apply Customer Merge* SQL task executes the batch procedure if rows were staged. That procedure applies Type 0/1 customer rules and Type 2 demographic history in a transaction:

```
1 DECLARE @BatchID UNIQUEIDENTIFIER = CONVERT(UNIQUEIDENTIFIER, ?);
```

| Derived Column Name | Derived Column | Expression | Data Type | Length |
|---|---|---|---|---|
| CustomerKey_Str | <add as new column> | (DT_WSTR,20)CustomerKey | Unicode string [DT_WSTR] | 20 |
| BatchID | <add as new column> | (DT_GUID)("{" + @[User::BatchID] + "}") | unique identifier [DT_GUID] | |
| MaritalStatus | Replace 'MaritalStatus' | ISNULL(MaritalStatus) ? "U" : (UPPER(TRIM(MaritalStatus)) == "M" \|\| UPPER(TRIM(MaritalStatu... | Unicode string [DT_WSTR] | 1 |
| YearlyIncome | Replace 'YearlyIncome' | ISNULL(YearlyIncome) \|\| TRIM(YearlyIncome) == "" ? "Unknown" : TRIM(YearlyIncome) | Unicode string [DT_WSTR] | 30 |
| Gender | Replace 'Gender' | ISNULL(Gender) ? "U" : (UPPER(TRIM(Gender)) == "M" \|\| UPPER(TRIM(Gender)) == "F" ? UPP... | Unicode string [DT_WSTR] | 1 |
| CountryName | Replace 'CountryName' | ISNULL(CountryName) \|\| TRIM(CountryName) == "" ? "Unknown" : TRIM(CountryName) | Unicode string [DT_WSTR] | 50 |
| RegionGroup | Replace 'RegionGroup' | ISNULL(RegionGroup) \|\| TRIM(RegionGroup) == "" ? "Unknown" : TRIM(RegionGroup) | Unicode string [DT_WSTR] | 50 |
| TotalChildren | Replace 'TotalChildren' | ISNULL(TotalChildren) \|\| TotalChildren < 0 ? 0 : TotalChildren | four-byte signed integer [DT_I4] | |
| Education | Replace 'Education' | ISNULL(Education) \|\| TRIM(Education) == "" ? "Unknown" : TRIM(Education) | Unicode string [DT_WSTR] | 30 |
| Occupation | Replace 'Occupation' | ISNULL(Occupation) \|\| TRIM(Occupation) == "" ? "Unknown" : TRIM(Occupation) | Unicode string [DT_WSTR] | 30 |
| HomeOwnerFlag | Replace 'HomeOwnerF... | ISNULL(HomeOwnerFlag) ? (DT_UI1)0 : (HomeOwnerFlag ? (DT_UI1)1 : (DT_UI1)0) | Boolean [DT_BOOL] | |
| NumberCarsOwned | Replace 'NumberCarsO... | ISNULL(NumberCarsOwned) \|\| NumberCarsOwned < 0 ? 0 : NumberCarsOwned | four-byte signed integer [DT_I4] | |
| BirthDate | Replace 'BirthDate' | ISNULL(BirthDate) ? (DT_DBDATE)"1900-01-01" : BirthDate | database date [DT_DBDATE] | |

Figure 4.4: Derived Column: input data cleaning step.

```
IF EXISTS (
    SELECT 1
    FROM stg.CustomerProfile
    WHERE BatchID = @BatchID
)
BEGIN
    EXEC dbo.usp_ApplyCustomerProfileBatch @BatchID = @BatchID;
END;
```

Listing 4.4: Merge

### 4.2.2 RFM Live

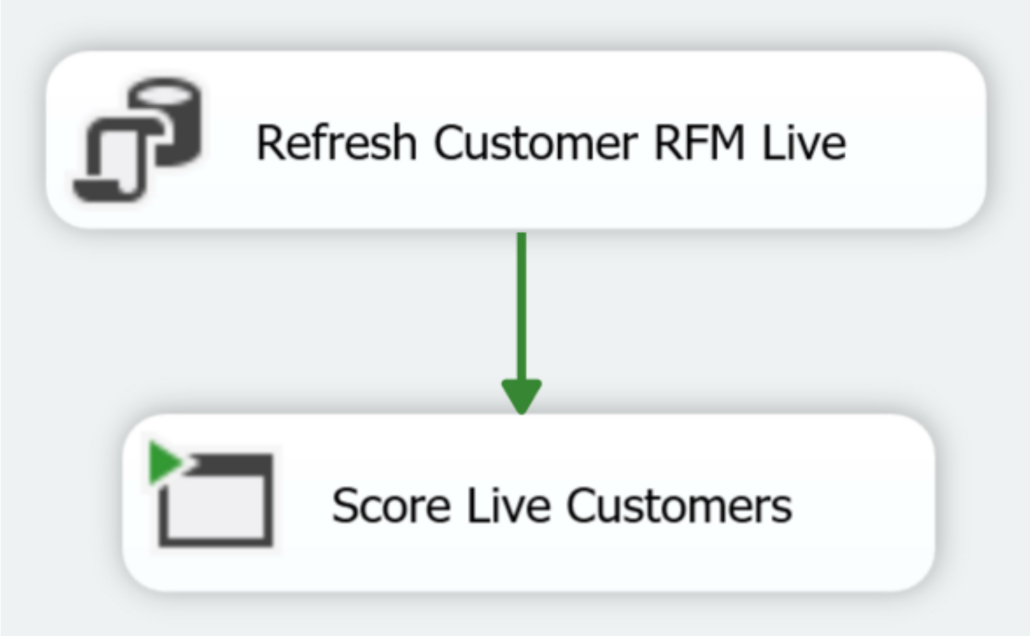



Figure 4.5: FactCustomerRFMLive handling.

The first SSIS task, *Refresh Customer RFM Live*, calls `usp_RefreshCustomerRFMLive`. A database watermark tracks the last processed order modification timestamp. When new or updated sales transactions are detected, the procedure recalculates order frequency and monetary totals for the affected customers and merges the results into `FactCustomerRFMLive`.

The second task, *Score Live Customers*, runs the Python scoring script `live_clustering.py`. It reads the latest active cluster centroids from `DimModelConfig`, standardizes the customer's RFM vector, and updates the segment label. Scheduled to run at one-minute intervals, this micro-batch pipeline delivers near-real-time customer segmentation.

### 4.2.3 RFM Snapshot

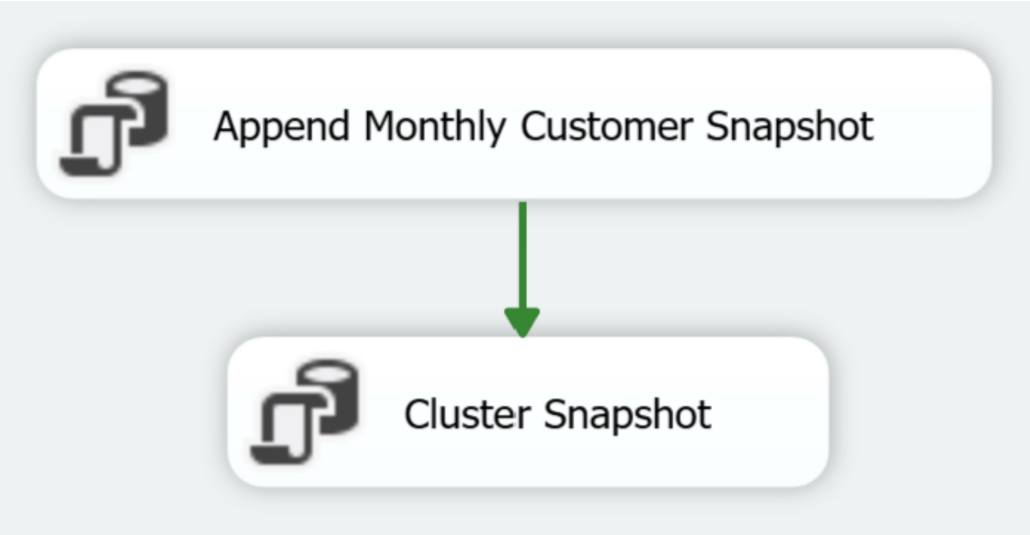



Figure 4.6: FactCustomerSnapshot handling.

The first task, *Append Monthly Customer Snapshot*, determines the previous month-end and calculates historical RFM values for the **FactCustomerSnapshot**:

```
DECLARE @SnapshotDate DATE;

SET @SnapshotDate = EOMONTH(GETDATE(), -1);

EXEC dbo.usp_AppendMonthlyRFMSnapshot
    @SnapshotDate = @SnapshotDate;
```

Listing 4.5: Monthly RFM Calculation

The subsequent *Cluster Snapshot* task assigns segment labels using the available effective-dated model definitions:

```sql
EXEC dbo.usp_ClusterCustomerRFMSnapshot;
```

Listing 4.6: Monthly Cluster Segment

## 4.3 Annual Model Training and Scheduling

The annual retraining workflow utilizes the year-end customer snapshot as training data to recalibrate cluster centroids. Following the initial 2011 bootstrap model, models for 2012 and 2013 were trained using `train_model.py`, with the resulting centroids and scaling parameters stored into `DimModelConfig`.

To automate future retraining, the Apache Airflow DAG `rfm_year_end_training` is configured to trigger on 1 January following the completion of the December snapshot, ensuring seamless annual model updates.

## 4.4 Pipeline Outputs

In summary, the automated ETL pipeline produces:

1. Current demographic profiles in `DimCustomerProfiling` (Type 1) and historical version tracking in `DimClusterProfiling` (Type 2);
2. Near-real-time operational segmentation in `FactCustomerRFMLive` triggered by incoming transactions;
3. Comprehensive monthly historical records in `FactCustomerSnapshot` for longitudinal analysis;
4. Versioned machine learning configurations in `DimModelConfig`.

This architecture provides the complete data foundation for the Power BI dashboards discussed in Chapter 6.

# Chapter 5

# RFM Analysis and K-Means Clustering

## 5.1 Motivation

The CompanyX transactional database contains purchases from May 2011 through June 2014. However, raw sales records alone do not identify which customer groups require different business strategies. This chapter follows the team's implemented approach: summarize purchase behaviour using RFM, clean and transform the features, fit K-Means to customer snapshots, and translate the resulting numeric clusters into interpretable segment names. The trained configurations can then be reused by the operational scoring pipeline and historical snapshot process.

### 5.1.1 RFM Variables

Recency (R) is the number of days from an evaluation date to the customer's last purchase. Frequency (F) is the count of distinct sales orders within the trailing 365 days. Monetary (M) is the sum of order-line amounts within that same 365-day period. Recency may exceed 365 days for customers who have not purchased recently. The year-end training snapshots used by the project are 31 December 2011, 2012, and 2013; there is no 2014 trained model.

## 5.2 Data Cleaning

The raw data is first ingested from the FactCustomerSnapshot, and then processed through deduplication. After that, any R, F, or M values are missing, the process stops and reports the number of affected rows to the ETL Team. Anomalies values such as negative values are set to zero, since a purchase count or a monetary value can not be negative.

This subset of data serves a representative overview of customer purchases based on RFM analysis as mentioned above.

| Customer | Recency | Frequency | Monetary |
|---|---|---|---|
| 11057 | 37 | 1 | 3729.36 |
| 11058 | 22 | 1 | 3756.99 |
| 11060 | 12 | 1 | 3756.99 |
| 11061 | 18 | 1 | 3729.36 |
| 11072 | 0 | 1 | 3756.99 |
| 11238 | 84 | 1 | 3756.99 |
| 11239 | 71 | 1 | 3729.36 |
| 11240 | 32 | 1 | 3756.99 |
| 11243 | 38 | 1 | 3756.99 |
| 11244 | 47 | 1 | 3756.99 |
| 11247 | 41 | 1 | 3729.36 |
| 11250 | 14 | 1 | 3756.99 |
| 11333 | 10 | 1 | 3756.99 |
| 11334 | 27 | 1 | 3756.99 |
| 11337 | 22 | 1 | 3729.36 |
| 11591 | 79 | 1 | 3953.99 |
| 11592 | 79 | 1 | 3953.99 |
| 11593 | 69 | 1 | 3953.99 |
| 11599 | 72 | 1 | 3953.99 |
| 11601 | 71 | 1 | 3953.99 |
| 11606 | 82 | 1 | 3953.99 |
| 11607 | 71 | 1 | 3953.99 |

Table 5.1: Example RFM records before preprocessing.

## 5.3 Log transformation

Spending and purchase counts are usually right-skewed. Most customers spend a moderate amount, while only a small number spend many times more. If raw values are used, these large values stretch the scale and the K-Means model spends more effort unnecessarily separating a few high spenders (outliers) rather than grouping typical customers. By using log transformation, the issue can be minimized.

For example, a customer who spends 3,000 and 300,000 differs by a factor of 100 in raw terms, but their log values differ by only 4.6. We use log(1+x) in this case, written log1p because some customers have a value of zero, which leads to log(0) undefined.

| Customer | Recency | Frequency | Monetary |
|---|---|---|---|
| 11057 | 3.638 | 0.693 | 8.224 |
| 11058 | 3.135 | 0.693 | 8.232 |
| 11060 | 2.565 | 0.693 | 8.232 |
| 11061 | 2.944 | 0.693 | 8.224 |
| 11072 | 0.000 | 0.693 | 8.232 |
| 11238 | 4.443 | 0.693 | 8.232 |
| 11239 | 4.277 | 0.693 | 8.224 |
| 11240 | 3.497 | 0.693 | 8.232 |
| 11243 | 3.664 | 0.693 | 8.232 |
| 11244 | 3.871 | 0.693 | 8.232 |
| 11247 | 3.738 | 0.693 | 8.224 |
| 11250 | 2.708 | 0.693 | 8.232 |
| 11333 | 2.398 | 0.693 | 8.232 |
| 11334 | 3.332 | 0.693 | 8.232 |
| 11337 | 3.135 | 0.693 | 8.224 |
| 11591 | 4.382 | 0.693 | 8.283 |
| 11592 | 4.382 | 0.693 | 8.283 |
| 11593 | 4.248 | 0.693 | 8.283 |
| 11599 | 4.290 | 0.693 | 8.283 |
| 11601 | 4.277 | 0.693 | 8.283 |
| 11606 | 4.419 | 0.693 | 8.283 |
| 11607 | 4.277 | 0.693 | 8.283 |

Table 5.2: Log-transformed sample RFM values.

## 5.4 Standardization

Before importing data to run the K-Means Clustering model, it is essential to standardize the RFM values to ensure that each metric contributes equally to the analysis, since Recency, Frequency, and Monetary values are measured on different scales. K-Means relies heavily on distances (later will be shown), so a variable with a larger spread would dominate the result.

StandardScaler solves this by subtracting the mean and dividing by the standard deviation of each variable, so that each one has a mean of 0 and a standard deviation of 1. The mean and standard deviation are estimated from the training data and stored, so that new customers are scaled in the same way.

| Customer | Recency | Frequency | Monetary |
|---|---|---|---|
| 11057 | 0.109 | 0.000 | -0.875 |
| 11058 | -0.392 | 0.000 | -0.581 |
| 11060 | -0.962 | 0.000 | -0.581 |
| 11061 | -0.583 | 0.000 | -0.875 |
| 11072 | -3.522 | 0.000 | -0.581 |
| 11238 | 0.913 | 0.000 | -0.581 |
| 11239 | 0.747 | 0.000 | -0.875 |
| 11240 | -0.032 | 0.000 | -0.581 |
| 11243 | 0.135 | 0.000 | -0.581 |
| 11244 | 0.342 | 0.000 | -0.581 |
| 11247 | 0.209 | 0.000 | -0.875 |
| 11250 | -0.819 | 0.000 | -0.581 |
| 11333 | -1.128 | 0.000 | -0.581 |
| 11334 | -0.196 | 0.000 | -0.581 |
| 11337 | -0.392 | 0.000 | -0.875 |
| 11591 | 0.852 | 0.000 | 1.454 |
| 11592 | 0.852 | 0.000 | 1.454 |
| 11593 | 0.719 | 0.000 | 1.454 |
| 11599 | 0.761 | 0.000 | 1.454 |
| 11601 | 0.747 | 0.000 | 1.454 |
| 11606 | 0.889 | 0.000 | 1.454 |
| 11607 | 0.747 | 0.000 | 1.454 |

Table 5.3: Standardized sample RFM values.

## 5.5 K-Means Clustering

K-Means is an unsupervised algorithm that partitions observations into $K$ clusters by iteratively assigning each point to the nearest centroid and updating centroid locations. The project fits scikit-learn `KMeans` with $K = 5$, `k-means++` initialization, `n_init=10`, `max_iter=300`, and `random_state=42`.

For each data point in the dataset, calculate the Euclidean distance between the point and each centroid. Assign the data point to the cluster with the closest centroid

$$d(\mathbf{x}, \mathbf{c}) = \sqrt{\sum_{j=1}^{3}(x_j - c_j)^2}.$$

Where:

- $d(x, y)$: the distance between data point x and centroid y
- $y_i$: donates the value of the i-th attribute value of centroid y
- $x_i$: represents the i-th attribute of the data point x
- $i$ : refers to index of the attribute
- $n$ : indicates the total number of the attribute for each data point

Within-cluster sum of squares (WCSS), also called inertia, measures the total squared distance from observations to their assigned centroids:

$$\text{WCSS} = \sum_{k=1}^{K}\sum_{i \in S_k}\sum_{j=1}^{3}(x_{ij} - c_{kj})^2.$$

Where:

- K: Total number of clusters.
- k: An index that refers to the current cluster, ranging from 1 to K.
- $S_k$: The set of data points assigned to cluster k.
- p: Number of features or dimensions in the dataset.
- j: An index that refers to the current feature or dimension, ranging from 1 to p.
- $x_{ij}$: The j-th feature value of the i-th data point.
- $\bar{x}_{kj}$: The mean value of the j-th feature in cluster k (i.e., the centroid's value).

- $(x_{ij} - \bar{x}_{kj})^2$: The squared Euclidean distance between the j-th feature of data point xi and the corresponding feature of the centroid of its cluster.

This WCSS can also be known as the Inertia, the lower means the tighter the data points are grouped. Another important criteria, which is Silhouette, also needs to be considered. The silhouette score measures how well-separated and cohesive clusters are in K-Means clustering. It evaluates each data point's fit relative to its own cluster versus neighboring clusters.

Figures 5.1 and 5.2 reproduce the diagnostic plots already saved with the project for the 31 December 2013 snapshot. Inertia decreases as $K$ increases, while the silhouette metric measures separation between clusters. The silhouette score for $K = 5$ is approximately 0.518; $K = 2$ achieves a higher score (approximately 0.636) on this snapshot. Therefore five clusters were chosen as a *business-interpretability trade-off*, supported by the elbow/silhouette diagnostics and the desire for distinct actionable customer profiles, not because five globally maximizes the silhouette score.

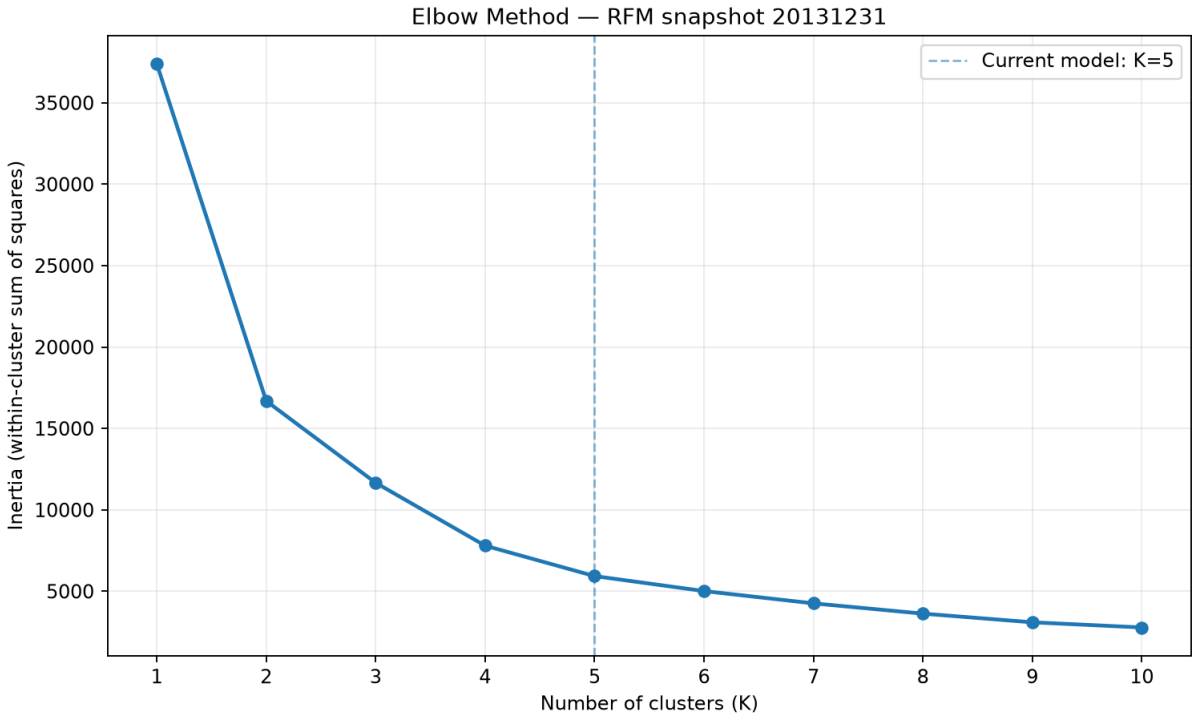



Figure 5.1: Elbow plot for the 31 December 2013 RFM snapshot (existing project diagnostic).

K-Means only returns a numeric cluster ID (0 to 4) for each customer, which reflects no value of what the groups mean. Moreover, annually the model is retrained and the number from 0 to 4 is different and not staying consistent. Therefore, to turn them into meaningful clusters automatically, a solution is implemented: Each cluster on an RFM variable is given "points", based on how its average compares with the other clusters. The rules are

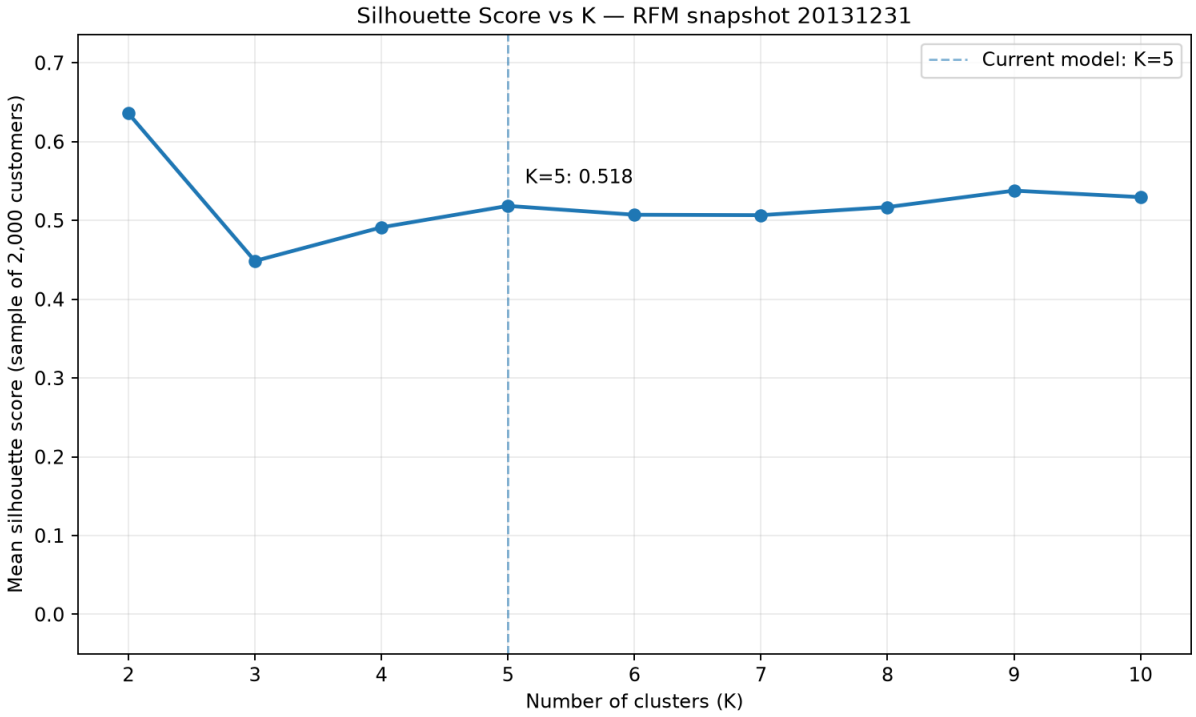



Figure 5.2: Silhouette comparison for different cluster counts on the same snapshot.

For Monetary and Frequency: the cluster with the highest average gets 5 points and the lowest get 1. (the higher these values are, the more valuable they are)

For Recency: the rule is reversed because shorter time since the last purchase is better. The implemented composite score is Monetary rank + Frequency rank + 1.1 × Recency rank.

Recency is weighted slightly higher, because a recent purchase is a strong sign that a customer is still active. Points are given to clusters, not to individual customers.

The outcome will look like this:

| Cluster | Avg Recency | Avg Frequency | Avg Monetary | R points | F points | M points | Score | Segment |
|---|---|---|---|---|---|---|---|---|
| 4 | 110.34 | 2.55 | 1,694.00 | 4 | 5 | 5 | 14.4 | Champions |
| 2 | 13.31 | 1.24 | 96.10 | 5 | 4 | 3 | 12.5 | Recent / Promising |
| 0 | 174.17 | 1.00 | 1,518.38 | 3 | 2 | 4 | 9.3 | High-Value / Cooling Off |
| 3 | 196.03 | 1.03 | 48.91 | 2 | 3 | 2 | 7.2 | Low-Spend / Occasional |
| 1 | 602.92 | 0.00 | 0.00 | 1 | 1 | 1 | 3.1 | Lost / Dormant |

Table 5.4: RFM Cluster Statistics and Segment Assignment

The example model output was recorded for an 18,484-customer analysis extract. The final dashboard chapter reports 18,485 customers for its June 2014 view. This one-record difference has not been reconciled in the submitted materials, so the two counts should not be presented as an identical population without verification.

Champions (cluster 4) has the highest frequency and spend, with an average of 2.55 purchases

and 1,694 in total. Its average recency is 110 days, so these customers are valuable but have not bought recently.

High-Value / Cooling Off (cluster 0) is the second-largest group by spend, but each customer has bought only once on average. Their last purchase was about 174 days ago, so they are at risk of leaving.

Low-Spend / Occasional (cluster 3) is the largest group. Customers buy about once, spend about 49 on average, and last bought about 196 days ago.

Recent / Promising (cluster 2) bought most recently, with an average of 13 days since the last purchase. Their spend is low so far, which suggests they are new customers who may grow.

Lost / Dormant (cluster 1) has no recorded frequency or spend in the period, and an average recency of 603 days. These customers have stopped buying.

## 5.6 Conclusion

The result of the RFM Analysis and K-Means clustering reveal the distinct customer segments with unique purchasing behaviours. By standardizing the RFM variables and applying the K-Means algorithm, the dataset was successfully divided into five interpretable clusters. Such segmentation highlights not only the diversity in customer engagement levels but also provides valuable insights for strategic business decision-making. Therefore, the CompanyX will be able to design and launch effective marketing strategies to each cluster of customers, not only to retain the occasional buyers, but also to come up with attractive promotions for recent customers.

The first historical model was initialized with `bootstrap.py` from the 2011 year-end snapshot, including retrospective labels for earlier months. Separate 2012 and 2013 models were then saved with `train_model.py`. The most recent active model is used by the Python live scorer. These saved model versions support repeatable clustering without retraining on every purchase.

# Chapter 6

# Business Intelligence & Analytics Dashboards

## 6.1 Context & Dashboard Objectives

### 6.1.1 Business Context

Following the construction of the dimensional data warehouse and the K-Means clustering model, our team built an interactive Business Intelligence dashboard using Microsoft Power BI. While underlying SQL tables store comprehensive transaction records and cluster assignments, business managers require intuitive visual summaries to track performance and guide operational decisions.

The dashboard connects directly to the data warehouse, utilizing the monthly snapshot slice at `SnapshotDateKey = 20140630` as its primary analytical reporting dataset. This fixed cutoff date provides a standardized baseline to compare recency, spending, and inactivity across the entire customer base. The Power BI report is organized into three interconnected pages:

1. **Executive Overview:** Enables senior directors to monitor company-wide customer volume, total revenue, longitudinal segment migration, and customer inactivity alerts.

2. **Customer Demographics:** Assists marketing and sales teams in profiling customers by geographic location, income bracket, age distribution, gender, and occupation.

3. **RFM Segmentation and Behavior:** Provides CRM managers with detailed behavioral metrics (Recency, Frequency, Monetary) across the five clusters to design targeted retention strategies.

### 6.1.2 Visualization Principles

To ensure clarity, visual hierarchy, and intuitive navigation, the dashboard adheres to established data visualization standards (Wilke 2019):

- **Treemaps for Categorical Hierarchies:** Used for multi-category distributions such as customer occupations, offering superior readability compared to crowded pie charts.
- **100% Stacked Column Charts for Proportions Over Time:** Applied to illustrate the structural evolution of customer segments across the four operational years (2011–2014).
- **Binned Histograms for Demographic Spans:** Customer ages are grouped into distinct demographic intervals ($< 30$, $30 - 40$, $41 - 50$, $51 - 60$, $60+$) to reveal primary customer life stages.
- **Bubble Charts for Multi-Dimensional Relationships:** Plotted with Recency on the X-axis, Monetary spend on the Y-axis, customer volume as bubble size, and benchmark reference lines to segment buyers into four strategic quadrants.
- **Consistent Visual Palette:** Segment colors remain strictly synchronized across all three pages to prevent cognitive friction:
  - **1. Champions:** Crimson Red (#A82020)
  - **2. High-Value / Cooling Off:** Forest Green (#22593D)
  - **3. Recent / Promising:** Vibrant Orange (#D97F1A)
  - **4. Low-Spend / Occasional:** Deep Teal (#259696)
  - **5. Lost / Dormant:** Royal Purple (#4C2773)

### 6.1.3 Dashboard Structure

The report follows a top-down analytical hierarchy:

1. **Page 1: Executive Customer & Segmentation Overview** (Macro business health and longitudinal trends);
2. **Page 2: Customer Demographics & Persona** (Detailed customer attributes and slicer drill-downs);
3. **Page 3: Customer RFM Segmentation & Behavior** (Cluster metrics and strategic positioning matrix).

## 6.2 Page 1: Executive Customer & Segmentation Overview

Page 1 provides senior executives with a high-level cockpit summarizing customer health, spending volume, and retention risks, as shown in Figure 6.1.

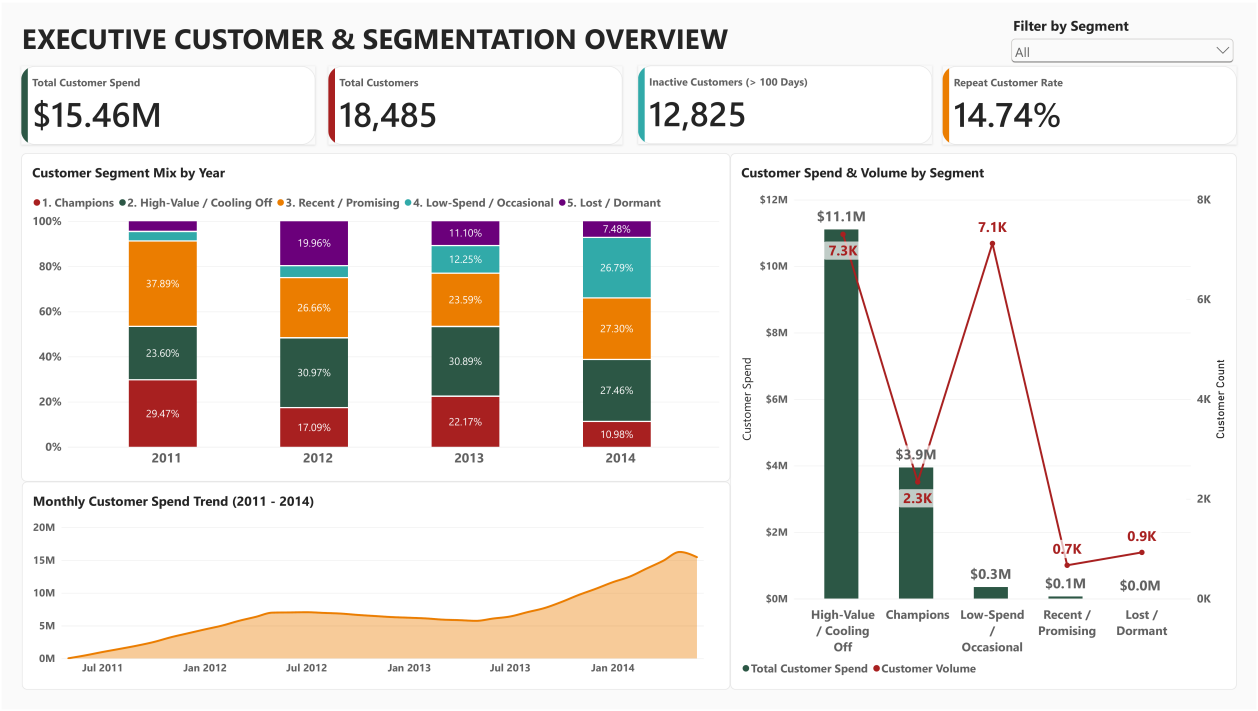



Figure 6.1: Executive Customer & Segmentation Overview Dashboard (Page 1).

### 6.2.1 Core Executive KPIs

The top ribbon highlights four primary performance indicators:

1. **Total Customer Spend ($15.46M):** The total customer monetary value recorded in the warehouse over the trailing 365-day observation window ($15,455,666.27).

2. **Total Customers (18,485):** The distinct count of individual retail customers captured in the reporting view as of 30 June 2014.

3. **Inactive Customers (> 100 Days) (12,825):** Customers whose recency exceeds 100 days without a purchase, representing **69.38%** of the customer base. Because bicycles and high-end gear are durable products, shoppers do not purchase monthly; however, exceeding 100 days without buying consumables, accessories, or service signals immediate risk of churn.

4. **Repeat Customer Rate (14.74%):** The percentage of customers with two or more orders in the trailing 365 days:

$$\text{Repeat Customer Rate} = \frac{\sum[\text{Customers with Frequency} \geq 2]}{\text{Total Customers}} = \frac{2{,}724}{18{,}485} = 14.74\% \quad (6.1)$$

This metric demonstrates that 85.26% of active buyers placed only a single order during the year, highlighting strong commercial potential for re-engagement.

### 6.2.2 Visual Charts and Key Findings

**1. Customer Segment Mix by Year (2011–2014)**

The 100% Stacked Column Chart illustrates the annual composition of customer segments over the four operating years:

- **2011 Baseline:** In its initial operating year, CompanyX focused on premium bicycles. *Recent / Promising* buyers made up the largest group at **37.89%**, followed by *Champions* at **29.47%** and *High-Value / Cooling Off* at **23.60%**. The *Low-Spend / Occasional* and *Lost / Dormant* segments accounted for the remaining 9.04% combined.
- **2012 Maturation:** As initial buyers aged without repeat purchases, the share of *High-Value / Cooling Off* rose to **30.97%**, while *Recent / Promising* stood at **26.66%**. *Lost / Dormant* expanded to **19.96%**, and *Champions* represented **17.09%**.
- **2013 Product Diversification:** CompanyX expanded its accessory and apparel offerings in 2013. The share of *High-Value / Cooling Off* stabilized at **30.89%**, *Recent / Promising* was **23.59%**, and *Champions* accounted for **22.17%**. Concurrently, *Low-Spend / Occasional* buyers surged to **12.25%**, while *Lost / Dormant* was **11.10%**.
- **2014 Mass Market Expansion:** By 2014, the proportion of *Low-Spend / Occasional* shoppers reached **26.79%**. Concurrently, *High-Value / Cooling Off* accounted for **27.46%**, *Recent / Promising* stood at **27.30%**, *Champions* represented **10.98%**, and *Lost / Dormant* was **7.48%**.

These shifts demonstrate CompanyX's transition from a high-end bicycle specialist in 2011 into a diversified mass-market sports retailer by 2014.

**2. Monthly Customer Spend Trend (2011–2014)**

The Area Chart plots trailing 365-day customer spend sampled at monthly snapshot intervals:

- **Steady Build-Up (2011–Mid 2012):** Reflects steady early-stage growth as the initial customer foundation was established.

- **Rapid Acceleration (Mid 2012–Late 2013):** Rolling customer spend climbed rapidly throughout 2013, driven by catalog expansion and increasing order volumes.
- **Stabilization (Late 2013–2014):** Spending stabilized at a mature plateau near $15.5M annually.

**3. Customer Spend & Volume by Segment**

The Dual-Axis Combo Chart highlights spending concentration across clusters:

- **Extreme Revenue Concentration:**
  - The *High-Value / Cooling Off* segment accounts for **$11.10M** (71.83% of total spend) across 7,305 customers (39.52% of all buyers).
  - The *Champions* cluster contributes **$3.94M** (25.50% of total spend) from 2,339 customers (12.65%).
  - Combined, these two premium segments generate **97.33%** of company revenue while comprising **52.17%** of the customer base.
- **The Low-Spend Contrast:** In contrast, *Low-Spend / Occasional* buyers make up 38.52% of customers (7,121 individuals) but generate only **$0.35M** (2.25% of revenue), while *Recent / Promising* accounts for **$0.06M** (0.41%) and *Lost / Dormant* contributes **$0.00**.
- **Strategic Takeaway:** Marketing investments must prioritize retaining high-value buyers, as losing a small fraction of cooling-off customers would severely impact total revenue.

## 6.3 Page 2: Customer Demographics & Persona Profiling

Page 2 helps us understand who our customers are, where they live, and what their personal characteristics look like.

### 6.3.1 Demographic Overview KPIs

The four cards on the left show customer averages:

- **Total Customers:** 18,485.
- **Total Customer Spend:** $15.46M.
- **Average Spend per Customer: $836.12**.
- **Average Order Value (AOV): $720.34**.

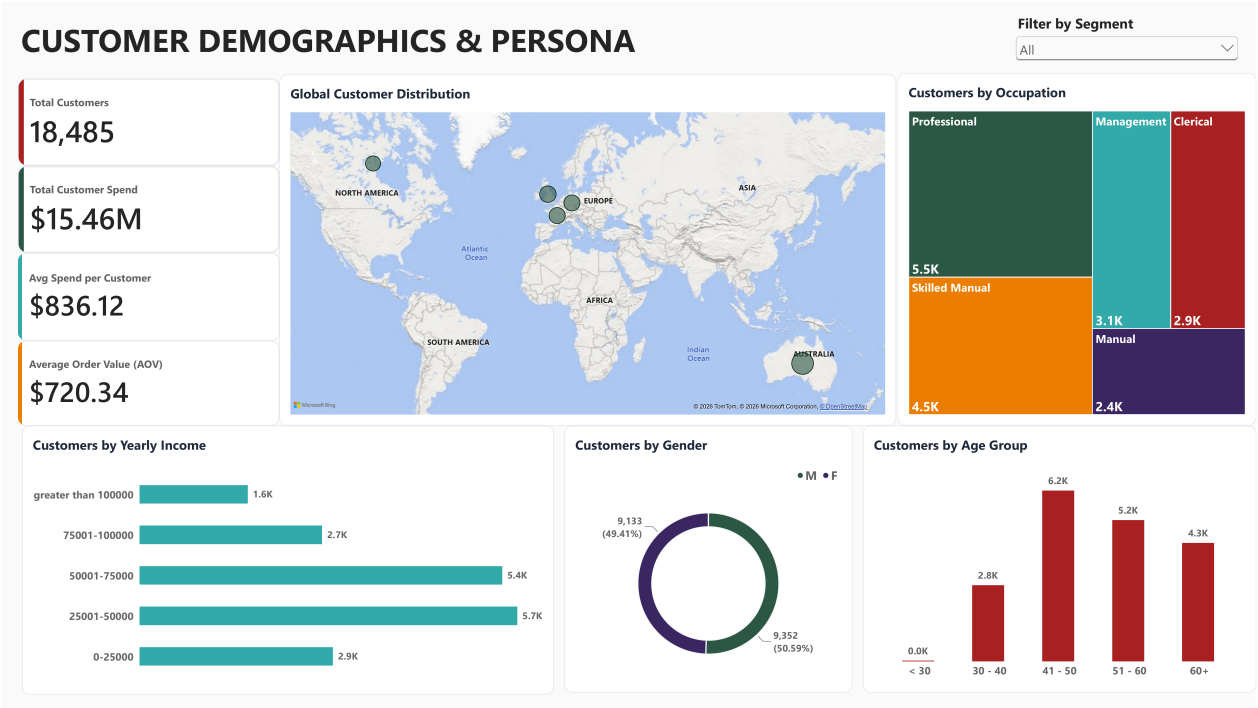



Figure 6.2: Customer Demographics & Persona Dashboard (Page 2).

These measures describe the same trailing 365-day period, and their ratio is consistent with comparatively low order frequency. They do not by themselves establish customers' lifetime order counts.

### 6.3.2 Demographic Analysis (Overall Base)

#### 1. Geographic Distribution (Map Visual)

The map shows that our customers live in developed markets with strong cycling cultures, mainly in North America (United States, Canada), Western Europe (United Kingdom, Germany, France), and Australia.

#### 2. Customers by Occupation (Treemap)

- **Professional** (5,505 people, 29.8%) and **Skilled Manual** (4,577 people, 24.8%) make up more than **54.6%** of all customers.
- Other occupations include Management (3,075, 16.6%), Clerical (2,928, 15.8%), and Manual workers (2,400, 13.0%).
- This shows that our core buyers are educated workers and skilled technicians who have good income and an active lifestyle.

**3. Customers by Yearly Income (Horizontal Bar Chart)**

- The middle-income groups are the largest: **$25,001–$50,000** has 5,720 customers (30.9%), and **$50,001–$75,000** has 5,496 customers (29.7%).
- Together, middle-class customers make up **60.6%** of the total customer base. High earners ($> \$100,000$) represent 1,600 customers (8.7%), while entry-level earners ($0–$25,000) account for 2,870 customers (15.5%).

**4. Customers by Gender (Donut Chart)**

- **Male:** 9,352 customers (50.59%).
- **Female:** 9,133 customers (49.41%).
- The gender ratio is almost 50/50, suggesting a balanced gender composition in the analyzed customer population.

**5. Customers by Age Group (Column Histogram)**

- Customers are grouped into age brackets: $< 30$, $30 - 40$, $41 - 50$, $51 - 60$, and 60+.
- The majority of customers are in middle age: **41–50 years old** (6,200 people, 33.5%) and **51–60 years old** (5,200 people, 28.1%).
- People under 30 make up a very small portion, within the age distribution captured by the dashboard; product-price preferences cannot be inferred from age alone.

### 6.3.3 Segment-Specific Demographic Personas (Slicer Drill-Down)

By utilizing the interactive segment slicer on Page 2, decision-makers can drill down into the unique demographic persona of each customer cluster. Table 6.1 provides an empirical comparison across the five groups.

Table 6.1: Comparative Demographic Profiles Across Customer Segments.

| Segment | Customers | Total Spend | AOV ($) | Top Occupation | Peak Income | Dominant Age |
|---|---|---|---|---|---|---|
| 1. Champions | 2,339 | $3.94M | $661.88 | Professional (38.9%) | $50k–$75k (30.7%) | 41–50 (38.3%) |
| 2. High-Value | 7,305 | $11.10M | $1,519.72 | Professional (29.9%) | $50k–$75k (32.1%) | 41–50 (39.4%) |
| 3. Recent | 669 | $64.12K | $77.22 | Skilled Manual (28.7%) | $25k–$50k (29.8%) | 60+ (29.5%) |
| 4. Low-Spend | 7,121 | $348.55K | $47.35 | Professional (27.9%) | $25k–$50k (30.6%) | 60+ (29.3%) |
| 5. Lost / Dormant | 927 | $0.00 | $0.00 | Skilled Manual (37.8%) | $25k–$50k (42.6%) | 41–50 (39.6%) |

**Key Findings from Segment Drill-Down**

Filtering each segment on Page 2 highlights clear differences in purchasing power, age distribution, and professional background across the five clusters:

1. **Champions (Affluent White-Collar Repeat Buyers):**
   - *High Earning Power:* Champions have the highest income level among all clusters. Buyers earning more than $100,000 account for **14.02%** (328 individuals), while **30.74%** (719 individuals) earn between $50,000 and $75,000.
   - *White-Collar Dominance: Professional* (38.86%, 909 people) and *Management* (18.98%, 444 people) make up **57.85%** of this group, showing that corporate earners drive repeat high-value purchases.
   - *Distinct Repeat Ordering:* This is the only segment where Average Spend per Customer ($1,685.11) is substantially higher than Average Order Value (**$661.88**), reflecting sustained repeat orders ($F \approx 2.55$).
   - *Age Concentration:* Heavily concentrated in the **41–50** age bracket (**38.26%**, 895 buyers), followed by 51–60 (**27.92%**, 653 buyers). These are established professionals investing in premium cycling and health.
2. **High-Value / Cooling Off (Single-Purchase Premium Buyers):**
   - *High Initial Basket Size:* Average Spend per Customer exactly equals Average Order Value at **$1,519.72**, indicating that each customer made a single major bicycle purchase ($F = 1.00$) approximately 5.8 months ago ($R = 174.18$ days) and has not returned.
   - *Middle-Class Core:* The primary income brackets are $50,001–$75,000 (32.11%, 2,346 buyers) and $25,001–$50,000 (31.62%, 2,310 buyers), representing **63.74%** of the cluster.
   - *Age Demographic:* Dominated by mature adults aged 41–50 (39.36%, 2.9K buyers) and 51–60 (28.55%, 2.1K buyers).
3. **Recent / Promising (Senior Entrants & Skilled Technicians):**
   - *Senior Age Demographic:* The single largest age bracket in *Recent / Promising* is **60+ years old** (**29.46%**, 200 customers), highlighting senior cyclists and retirees as an emerging acquisition segment.
   - *Occupational Profile: Skilled Manual* workers lead at **28.72%** (195 people), followed closely by *Professional* at 27.54%.

- *Early Repeat Signals:* With an AOV of \$77.22 and an average spend of \$95.85, a subset of customers have already completed second accessory orders within two weeks of their first purchase ($R = 13.24$ days).

4. **Low-Spend / Occasional (Budget-Conscious Accessory Shoppers):**
   - *Lower Income Skew:* Contains the largest share of entry-level earners, with buyers earning below \$25,000 accounting for **20.57%** (1,465 individuals).
   - *Occupational Spread: Manual* workers represent 16.7% (1.2K people), the highest proportion among all active clusters.
   - *Age Profile:* Shows a bi-modal distribution split between seniors aged **60+** (**29.3%**) and young adults aged 30–40 (18.1%).
   - *Low Basket Size:* An average order value of **\$47.35** indicates frequent purchases of low-cost consumables and basic replacement parts.
5. **Lost / Dormant (Disengaged Lower-Income Buyers):**
   - *Occupational Composition: Skilled Manual* workers heavily dominate at **37.76%** (350 buyers), while *Management* drops to **9.17%** (85 buyers).
   - *Income Distribution:* Over **88.67%** of members (822 buyers) earn under \$75,000 annually, with **42.61%** (395 buyers) in the \$25,001–\$50,000 bracket.
   - *Prolonged Inactivity:* With \$0.00 in spend and an average recency of 608.69 days, these customers have completely stopped purchasing.

## 6.4 Page 3: Customer RFM Segmentation & Behavioral Matrix

Page 3 explains the 5 customer clusters from our K-Means model using the RFM framework (Recency, Frequency, Monetary).

### 6.4.1 Company-Wide Benchmark Metrics

The top ribbon gives the averages for the 30 June 2014 reporting snapshot:

- **Average Recency: 190.57 Days** (around 6.3 months since the last order).
- **Average Frequency: 1.17 Orders** per customer.
- **Average Monetary: \$841.77** total spend per customer.

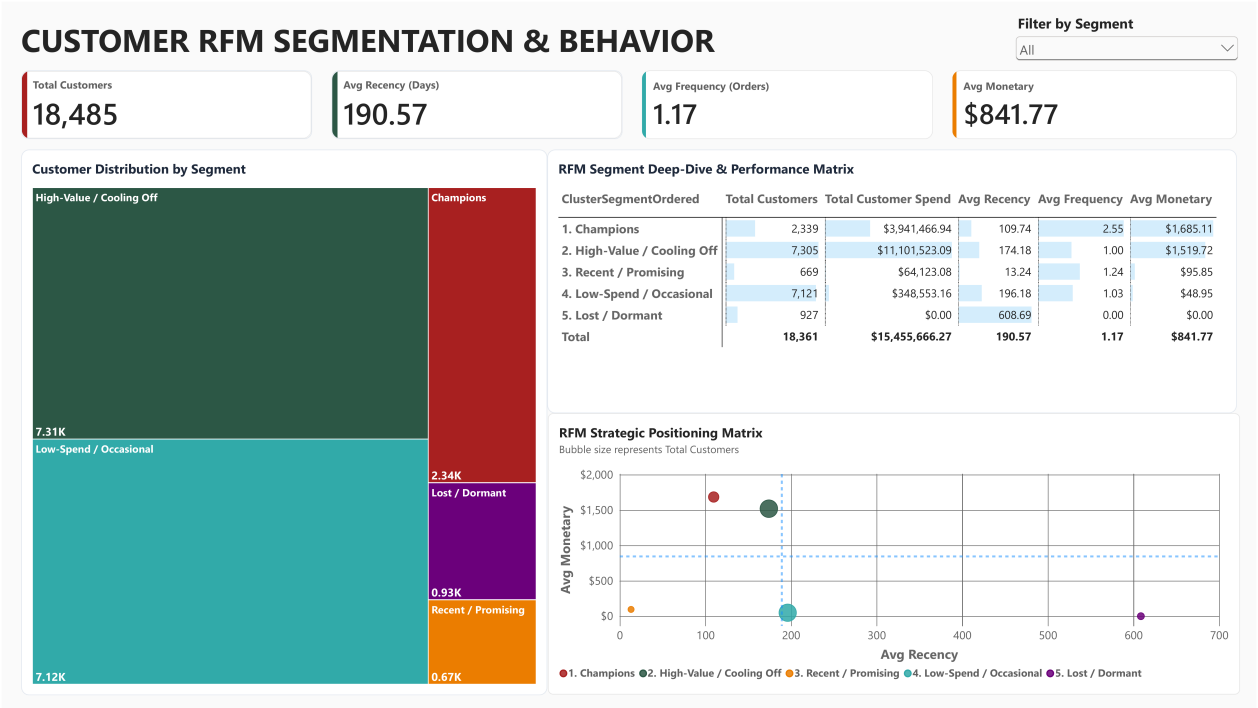



Figure 6.3: Customer RFM Segmentation & Behavior Dashboard (Page 3).

### 6.4.2 Cluster Performance Table

Table 6.2 reproduces the five segment statistics reported by the updated dashboard export (reflecting the exclusion of cancelled sales orders):

Table 6.2: Customer RFM Cluster Statistics.

| Cluster Segment | Customers | Total Spend ($) | Avg R (Days) | Avg F | Avg M ($) |
|---|---|---|---|---|---|
| 1. Champions | 2,339 (12.74%) | $3,941,466.94 | 109.74 | 2.55 | $1,685.11 |
| 2. High-Value / Cooling Off | 7,305 (39.79%) | $11,101,523.09 | 174.18 | 1.00 | $1,519.72 |
| 3. Recent / Promising | 669 (3.64%) | $64,123.08 | 13.24 | 1.24 | $95.85 |
| 4. Low-Spend / Occasional | 7,121 (38.78%) | $348,553.16 | 196.18 | 1.03 | $48.95 |
| 5. Lost / Dormant | 927 (5.05%) | $0.00 | 608.69 | 0.00 | $0.00 |
| **Company Total / Average** | **18,361** | **$15,455,666.27** | **190.57** | **1.17** | **$841.77** |

### 6.4.3 Customer Volume Distribution (Treemap)

The Treemap on the left side of Page 3 illustrates the relative share of each customer group:

- **High-Value / Cooling Off:** 7,305 customers (**39.79%** of total volume, labeled 7.31K).
- **Low-Spend / Occasional:** 7,121 customers (**38.78%**, labeled 7.12K).

- **Champions:** 2,339 customers (**12.74%**, labeled 2.34K).
- **Lost / Dormant:** 927 customers (**5.05%**, labeled 0.93K).
- **Recent / Promising:** 669 customers (**3.64%**, labeled 0.67K).

This chart emphasizes that more than 78% of the customer base is divided almost evenly between two contrasting groups: high-spending buyers who need urgent retention, and low-spending buyers who make infrequent accessory purchases.

### 6.4.4 Strategic Positioning Matrix (Bubble Chart)

The bubble chart at the bottom-right plots all 5 clusters:

- **X-Axis:** Average Recency (from 0 to 700 days). Lower is better.
- **Y-Axis:** Average Monetary (from \$0 to \$2,000). Higher is better.
- **Bubble Size:** Total number of customers in the cluster.
- **Benchmark Lines:** Dotted reference lines at company averages ($X = 190.57$ days and $Y = \$841.77$) divide the chart into four quadrants.

**Quadrant Analysis**

1. **High spend, below-average recency — Champions:** These customers bought recently ($R = 109.74$ days) and spend the most ($M = \$1,685.11$). They are our best VIP customers.
2. **High spend, moderately recent — High-Value / Cooling Off:** These customers spent a lot ($M = \$1,519.72$), but their Recency is 174.18 days, which is close to the 190-day average. They are starting to cool down and need immediate attention before they leave.
3. **Low spend, very recent — Recent / Promising:** These are new customers who bought recently ($R = 13.24$ days). Their spend is still small (\$95.85), but they have great potential to grow into Champions.
4. **Low spend, long recency — Low-Spend and Lost:** These customers have not bought for a long time and spend very little. The Lost group has not returned for over 600 days ($R = 608.69$).

## 6.5 Summary of Core DAX Measures

Table 6.3 records the main measure *logic* described by the team. Names shown here are conceptual; the reporting model derives its table labelled `FactCustomerRFMLive` from the June 2014 snapshot, rather than from the SQL table of the same name. The final Power BI field expressions should be checked during dashboard handover:

Table 6.3: Dashboard measure definitions (conceptual DAX logic).

| **Measure Name** | **Home Table** | **DAX Formula / Logic** |
|---|---|---|
| Total Customers | FactCustomerRFMLive | `DISTINCTCOUNT(CustomerKey)` |
| Total Customer Spend | FactCustomerRFMLive | `SUM(Monetary)` |
| Inactive Customers (>100d) | FactCustomerRFMLive | `CALCULATE([Total Customers], Recency > 100)` |
| Repeat Customer Rate | FactCustomerRFMLive | `VAR R = CALCULATE([Total Customers], Frequency >= 2) RETURN DIVIDE(R, [Total Customers], 0)` |
| Average Order Value (AOV) | FactCustomerRFMLive | `DIVIDE([Total Customer Spend], SUM(Frequency), 0)` |
| Avg Spend per Customer | FactCustomerRFMLive | `DIVIDE([Total Customer Spend], [Total Customers], 0)` |
| Rolling 365-day Monetary Trend | FactCustomerSnapshot | `SUM(Monetary365d)` |

The business actions supported by these visualizations are discussed in Chapter 7.

# Chapter 7

# Business Insights and Decision-Making

## 7.1 From Customer Segments to Business Actions

This chapter bridges data analytics and commercial strategy. By combining executive trends, demographic personas, and RFM behavioral metrics, CompanyX can transition from generic, blanket promotions to highly targeted, cost-effective interventions.

Section 7.2 presents specific retention and growth strategies tailored to each of the five customer segments. Section 7.3 then outlines an end-to-end operational use case demonstrating how the Power BI dashboards directly guide marketing budget allocation and campaign evaluation.

## 7.2 Actionable Business Recommendations

Based on empirical insights from the dashboards, we recommend the following targeted strategies:

### 7.2.1 1. Champions: VIP Loyalty Program

- **Goal:** Keep our best customers happy and encourage them to recommend the brand to others.
- **Action:** Create a VIP club that gives early access to new bicycle models, invitations to special cycling events, and referral discounts when they introduce friends.

### 7.2.2 2. High-Value / Cooling Off: Urgent Retention Campaigns

- **Goal:** Protect a segment representing over 71% of the monetary amount in the final reporting snapshot.

- **Action:** Set up an automated email at day 120 after their purchase offering a free annual bicycle check-up at local partner shops, or a $150 discount voucher if they upgrade components within 30 days.

### 7.2.3 3. Recent / Promising: Cross-Selling and Welcome Series

- **Goal:** Encourage new buyers to make a second purchase quickly.
- **Action:** Send a 30-day email series: Day 3 with bike assembly and safety tips, Day 14 with cleaning advice, and Day 21 offering a 15% discount on essential accessories like helmets, gloves, and lights.

### 7.2.4 4. Low-Spend / Occasional: Low-Cost Automated Marketing

- **Goal:** Increase purchase frequency without spending too much marketing budget.
- **Action:** Avoid expensive phone calls or printed mail. Instead, send automated emails and seasonal clearance announcements for cheap consumables like water bottles and spare tubes.

### 7.2.5 5. Lost / Dormant: Cost Reduction

- **Goal:** Avoid costly broad advertising to a low-engagement segment (whose average snapshot recency exceeds 600 days).
- **Action:** Remove this group from paid Facebook and Google ads. Only send them one or two broad holiday sale emails per year (e.g., Black Friday or New Year).

## 7.3 Business Decision-Making Use Case

To demonstrate how the business intelligence dashboards directly support managerial decision-making, we present a concrete operational scenario facing the CompanyX marketing department at the end of the historical reporting period (Q3 2014).

### 7.3.1 Business Dilemma and Resource Constraints

In traditional retail marketing, companies often apply indiscriminate "blanket marketing"—distributing identical promotional materials to all registered customers. For CompanyX, distributing a generic email or paid advertising campaign across all 18,485 customers with a constrained quarterly marketing budget of $10,000 would result in an ineffective allocation of

approximately \$0.54 per customer. This dilutes capital, annoys dormant shoppers, and fails to offer meaningful incentives to high-value patrons.

Instead, the marketing director uses the Power BI analytical dashboards as a decision-support system to guide strategic resource allocation through an empirical four-step workflow.

### 7.3.2 Step-by-Step Decision Workflow via BI Dashboards

1. **Step 1: Macro-Risk Identification (Executive Overview — Page 1):** Examining the Executive Dashboard (Figure 6.1), the marketing director immediately observes two critical operational warnings:

   - *Severe Inactivity:* 12,825 customers (69.38% of the customer base) have not placed an order in over 100 days.
   - *Extreme Revenue Concentration:* The Dual-Axis Combo Chart reveals that \$11.10M (71.83% of trailing 365-day spend) is concentrated in the *High-Value / Cooling Off* cluster (7,305 customers), while *Champions* generate \$3.94M (25.50%).

   **Management Insight:** Over 97% of the company's financial inflow depends on high-spending buyers who have purchased only once on average and whose average recency has reached 174 days. If these buyers slip into complete dormancy, CompanyX faces an unsustainable revenue decline. Therefore, preventing churn in this group must take precedence over acquiring low-spend shoppers.

2. **Step 2: Audience Profiling via Slicers (Customer Demographics — Page 2):** To tailor the retention message, the campaign manager activates the *Filter by Segment* slicer on Page 2, filtering specifically for *2. High-Value / Cooling Off* (Table 6.1):

   - *Age and Income:* Buyers are predominantly mature adults aged 41–50 (39.42%) and 51–60 (28.55%), with 63.75% earning between \$25,000 and \$75,000 annually.
   - *Occupational Status: Professional* (29.9%) and *Skilled Manual* (25.1%) are the primary occupations.
   - *Spend Characteristics:* The Average Spend per Customer exactly equals the Average Order Value at \$1,519.72.

   **Management Insight:** These buyers are financially stable middle-class professionals who invested in a premium bicycle approximately 5.8 months ago. Because bicycles are durable products, these customers do not require a new bike; rather, their bicycle is approaching the standard 6-month threshold where routine maintenance, tire inspection, and component wear-and-tear become relevant.

3. **Step 3: Strategic Campaign Formulation and Budget Allocation:** Based on the RFM positioning matrix on Page 3 (Figure 6.3), the director divides the $10,000 budget into targeted initiatives rather than uniform spending:

   - **Primary Campaign (80% / $8,000) — "Rider Re-Activation Program":** Targets the 7,305 *High-Value / Cooling Off* customers. The company automates a personalized email dispatch at Day 120 offering:
     - A complimentary 12-point bicycle safety and tune-up voucher redeemable at certified partner retail workshops.
     - A time-limited $150 discount voucher for component or gear upgrades (e.g., performance saddles, hydraulic disc brakes, pedaling kits) valid on orders exceeding $300 within 30 days.

     *Unit Allocation:* Approximately $1.10 per targeted recipient (allocated across email marketing automation, redemption tracking, and dealer service reimbursement).
   - **Secondary Campaign (15% / $1,500) — "New Cyclist Welcome Series":** Targets the 669 *Recent / Promising* customers whose recency is just 13.24 days and average spend is $95.85.
     - A 3-step automated onboarding email sequence over 30 days providing bicycle assembly guides, local cycling route recommendations, and a 15% discount on essential accessories (helmets, lights, puncture repair kits) to stimulate their critical second transaction ($F \geq 2$).

     *Unit Allocation:* Approximately $2.24 per active contact.
   - **Cost-Avoidance Decision (5% / $500 Reserve + Overhead Savings):** The dashboard indicates that 927 customers in the *Lost / Dormant* cluster have generated $0.00 in spend and have an average recency exceeding 608 days. The marketing team immediately excludes this segment from ongoing paid digital advertising (e.g., Google Search and Meta Ads). This decision reclaims an estimated $2,500 in wasted annual ad spend, which is redirected to customer service enhancements.

4. **Step 4: Measurable Performance Indicators and ROI Estimation:** The success of these data-driven interventions is monitored directly on subsequent monthly snapshot dates using specific dashboard KPIs:

   - *Re-activation Rate:* Percentage of targeted High-Value buyers who redeem the maintenance voucher or place an accessory order within 60 days, successfully transitioning into the *Champions* cluster ($F \geq 2$). Even a conservative conversion rate of 3.0%

(219 reactivated buyers making an average follow-up purchase of $250) yields $54,750 in gross incremental sales, representing an estimated return on marketing investment (ROMI) of 584% on the $8,000 campaign spend.

- *Repeat Customer Rate:* Tracking whether the company-wide repeat rate increases from the baseline of 14.74%.
- *Customer Acquisition and Retention Efficiency:* Verifying that marketing cost per reactivated customer is substantially lower than new customer acquisition costs.

### 7.3.3 Methodological Boundaries and Analytical Rigor

This scenario demonstrates how the integration of dimensional modeling, RFM clustering, and Power BI dashboards transitions descriptive customer data into targeted, economically defensible business interventions.

It is important to emphasize that this section presents an analytical *decision-support use case* rather than an empirically executed campaign. In a commercial environment, validating the precise incremental revenue and causal lift would require conducting a randomized controlled trial (A/B testing), holding out an uncontacted control group against the treated customer segments.

**Interpretation note.** The dashboard's figures reflect the current saved report and reference-date filters. The ML chapter contains an illustrative 18,484-customer model extract, whereas this dashboard reports 18,485 customers. The team should reconcile that one-record discrepancy during the final dashboard refresh before asserting identical populations across the two exports.

# Chapter 8

# System Integration, Verification and Limitations

## 8.1 End-to-End System Integration

The completed Business Intelligence solution establishes an automated pipeline connecting operational transactions in `CompanyX` to an analytical data warehouse, predictive machine learning models, and interactive reporting dashboards.

The architecture balances operational responsiveness with historical depth:

- **Operational Path (`FactCustomerRFMLive`):** Incremental SSIS packages capture demographic modifications and sales transactions at one-minute intervals. The Python scoring engine applies active centroid parameters from `DimModelConfig` to reclassify affected buyers immediately upon purchase.
- **Historical Path (`FactCustomerSnapshot`):** Monthly scheduled routines compute standardized 365-day RFM snapshots as of each calendar month-end date, enabling longitudinal cohort analysis, inactivity tracking, and annual model recalibration.

## 8.2 Execution Schedule and Orchestration

The operational jobs run under automated schedules:

- **High-Frequency Micro-Batch (1-minute interval):** Deployed SSIS packages poll the operational database for changes using watermark timestamps: `Package_Customer_Incremental.dtsx` and `RFM_Live.dtsx`.
- **Monthly Snapshot (00:05 on the 1st of each month):** Package `RFM_Snapshot.dtsx` computes customer states for the preceding month-end.

- **Annual Model Retraining (1 January):** The Apache Airflow DAG `rfm_year_end_training` is configured to retrain cluster centroids on completed annual snapshots and register new model versions in `DimModelConfig`.

## 8.3 System Verification and Quality Assurance

To ensure reproducibility and analytical consistency, multiple automated and SQL-based verification checks were implemented:

1. **Demographic Integrity:** Verified that current customer counts in `DimCustomerProfiling` match source individual customer counts, and that historical profile changes are correctly versioned with effective date intervals in `DimClusterProfiling`.
2. **Snapshot Uniqueness:** Validated that each customer possesses at most one record per monthly snapshot date in `FactCustomerSnapshot`, with zero duplicate primary keys.
3. **Model Parameter Consistency:** Confirmed that each retrained model version saved in `DimModelConfig` contains exactly five cluster centroid definitions with valid scaling parameters and active date ranges.
4. **Data Reconciliation:** Provided the verification script `VERIFY_BUSINESS_FIGURES.sql` to audit key baseline metrics, including customer counts (18,485), aggregate spend ($15.46M), and cluster demographic statistics across the pipeline.

## 8.4 Operational Limitations and Future Considerations

- **Purchase-Time Recency Semantics:** By design, `FactCustomerRFMLive` sets recency to zero on purchase date. Consequently, tracking progressive customer inactivity requires consulting the monthly `FactCustomerSnapshot` table.
- **Airflow Sensor Dependency:** In a production environment, adding an explicit SQL sensor task to the Airflow DAG to verify that the SSIS December snapshot has fully committed before model training begins will ensure robust unattended execution.
- **Online Cloud Deployment:** The current system is successfully deployed in an on-premises development environment. Migrating the SQL Server warehouse and Power BI reports to Azure Synapse or Fabric represents an ideal next step for enterprise-scale distribution.
- **Empirical Campaign Validation:** While the dashboard provides actionable decision support, proving causal revenue lift requires executing live marketing campaigns with randomized holdout control groups.

## 8.5 Alignment with Project Requirements

The completed architecture directly fulfills all course specifications:

- **Data Warehouse & SCD:** A customer-centric fact constellation schema implementing SCD Types 0, 1, and 2;
- **ETL Pipeline:** Automated incremental extraction using SQL `EXCEPT` change detection and one-minute micro-batch scheduling;
- **Data Mining / Machine Learning:** RFM modeling, non-linear $\log(1 + x)$ transformations, z-score standardization, and K-Means clustering validated by Elbow and Silhouette metrics;
- **Business Intelligence Dashboard:** A three-page interactive Power BI report adhering to professional visual design standards;
- **Decision-Making Use Case:** A comprehensive four-step marketing resource allocation scenario with quantified ROMI projections.

# Chapter 9

# Conclusion and Future Work

This project delivers an end-to-end Business Intelligence and customer analytics solution for CompanyX, successfully bridging operational retail transactions with predictive machine learning and executive decision-making.

The core technical achievements include:

1. **Dimensional Warehouse Architecture:** Designed a customer-focused fact constellation schema featuring two complementary fact tables (`FactCustomerRFMLive` for near-real-time purchase events and `FactCustomerSnapshot` for standardized monthly observations) alongside SCD Type 0, 1, and 2 dimensions.

2. **Automated Incremental ETL Pipeline:** Deployed SSIS packages utilizing SQL `EXCEPT` change detection and database watermarks to execute micro-batch customer updates at one-minute intervals.

3. **Machine Learning Segmentation:** Implemented an unsupervised K-Means clustering model ($K = 5$) based on log-transformed and standardized RFM features, validated through quantitative Elbow and Silhouette analyses, and operationalized via centroid persistence in `DimModelConfig`.

4. **Interactive Business Intelligence Dashboards:** Built a three-page Power BI report providing executive health metrics, demographic persona drill-downs via slicers, and behavioral RFM quadrant analysis.

5. **Actionable Business Decision Support:** Formulated targeted marketing campaigns and demonstrated a quantified resource allocation use case yielding projected high-ROI customer retention.

## 9.1 Future Work

To further advance the platform toward enterprise-grade production, future development can focus on:

- **Cloud Migration and Scalability:** Transitioning the SQL Server warehouse and SSIS workflows to cloud-native platforms such as Microsoft Fabric, Azure Synapse, or Databricks for distributed data lakehouse storage;
- **Predictive Churn Modeling:** Augmenting the descriptive RFM segmentation with supervised churn prediction algorithms (e.g., XGBoost or Random Forests) to forecast probability of departure before customers surpass the 100-day inactivity threshold;
- **Automated Campaign Feedback Loops:** Integrating Power BI directly with customer relationship management (CRM) systems (e.g., Salesforce or HubSpot) to automate trigger emails and track real-time campaign redemption metrics.

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